

NYUv2 surface normals — d_l (%) ↑

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	22.22	24.55	23.66	32.84	13.23	36.31	16.88	0.53	4.57
AvgLayer	20.72	27.84	31.01	35.34	15.24	32.55	17.35	0.53	4.29
WeightedLayers	25.45	30.68	34.61	36.54	16.81	35.50	17.48	0.53	3.14
ConcatLayers	<u>30.91</u>	<u>35.21</u>	<u>39.26</u>	<u>41.00</u>	<u>24.71</u>	<u>39.57</u>	<u>22.46</u>	6.82	2.00
AMDF (ours)	35.91	40.75	45.19	46.96	29.01	45.48	28.38	1.62	1.00

NYUv2 depth — d_l (%) ↑

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	60.79	65.27	66.44	62.49	58.40	69.14	56.90	0.66	4.00
AvgLayer	53.39	63.70	65.50	65.82	61.21	66.92	58.95	0.66	4.57
WeightedLayers	57.60	65.21	66.66	66.45	61.27	68.53	59.03	0.66	3.43
ConcatLayers	65.10	70.72	73.21	69.28	66.36	<u>71.73</u>	61.42	6.95	1.14
AMDF (ours)	<u>62.42</u>	<u>69.30</u>	<u>71.83</u>	<u>68.23</u>	<u>63.84</u>	71.76	<u>59.46</u>	1.75	1.86

NYUv2 semantic segmentation — mIoU (%) ↑

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	<u>43.89</u>	53.38	<u>57.19</u>	39.08	40.85	<u>44.11</u>	32.88	0.54	3.21
AvgLayer	33.70	47.37	50.89	40.69	42.18	42.10	33.42	0.54	4.29
WeightedLayers	39.30	50.17	53.07	41.30	42.13	43.11	33.40	0.54	3.86
ConcatLayers	42.92	<u>51.46</u>	54.88	<u>41.44</u>	<u>43.36</u>	43.98	34.12	6.84	2.43
AMDF (ours)	44.04	53.38	57.20	42.05	43.39	45.26	<u>33.49</u>	1.64	1.21

Taskonomy — aggregate L₁ ×100 ↓

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	41.64	39.21	39.27	39.52	46.81	36.97	42.23	0.53	4.43
AvgLayer	43.66	39.42	38.50	38.55	45.51	37.84	41.82	0.53	4.43
WeightedLayers	41.46	38.48	37.49	38.14	44.40	37.27	41.68	0.53	3.14
ConcatLayers	37.76	<u>35.19</u>	<u>33.62</u>	<u>34.54</u>	<u>38.74</u>	33.45	<u>37.64</u>	6.82	1.71
AMDF (ours)	<u>38.82</u>	34.64	32.69	34.13	37.90	<u>35.10</u>	37.58	1.62	1.29

CUB-200 fine-grained classification — Top-1 (%) ↑

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
AvgLayer	41.19	60.29	54.71	37.41	81.45	29.96	41.39	0.63	6.29
WeightedLayers	64.19	77.83	73.01	45.30	81.11	33.60	43.43	0.63	4.71
ConcatLayers	56.92	73.40	71.67	49.83	81.71	<u>41.38</u>	42.72	6.92	4.57
AvgPool	62.70	78.19	71.79	25.58	79.66	27.05	39.81	0.20	5.86
EfficientProbe	77.22	84.01	86.19	74.38	<u>83.95</u>	74.81	71.94	1.27	1.14
AttentiveFusionAP	64.53	77.98	<u>77.51</u>	<u>50.17</u>	81.99	35.80	44.17	8.60	3.00
AMDF (ours)	<u>65.25</u>	<u>78.49</u>	75.23	47.13	84.62	35.88	<u>59.99</u>	1.72	2.43

NAVI camera pose — rotation (°) ↓ / translation ↓

Method	DINOv3-S		DINOv3-B		DINOv3-L		MAE-L		SigLIP2-L		V-JEPA2.1-L		PESpatial-L	
	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.
LastLayer	45.49	0.17	44.58	0.17	40.33	0.16	38.74	0.16	62.69	0.22	<u>37.05</u>	0.15	65.22	<u>0.23</u>
AvgLayer	43.97	0.17	36.13	0.15	35.44	0.14	36.73	<u>0.15</u>	59.33	<u>0.21</u>	47.21	0.18	66.16	<u>0.23</u>
WeightedLayers	43.31	0.17	35.28	0.15	35.06	0.14	34.88	0.14	<u>58.86</u>	<u>0.21</u>	43.40	<u>0.17</u>	67.01	0.24
ConcatLayers	36.71	0.15	30.30	0.13	28.37	0.12	31.74	0.14	57.31	0.20	37.58	0.15	57.66	0.21
AMDF (ours)	<u>41.40</u>	<u>0.16</u>	<u>32.14</u>	<u>0.14</u>	<u>32.80</u>	<u>0.13</u>	<u>32.17</u>	0.14	59.10	<u>0.21</u>	34.48	0.15	<u>60.31</u>	0.21

Bold red = best, underlined = second best per column. Rank = mean over backbone columns of the per-column rank (ties averaged; NAVI over 14 rot/trans sub-columns). Params (M) = trainable parameters (probe + task head) for ViT-L backbones (D=1024, 24 layers). Mean of seeds 8/42/123, best-epoch validation metric; frozen backbones; same training recipe for every row of a table. AMDF: depth-aware fusion, 16 expert queries, values projected to D/2, no bottleneck, no per-layer refinement.